Project Description

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Introduction:

In this report, there will be configurations of routers and a switch as part of the Layer 3 devices. In addition, there will be configurations of end devices such as PCs and Linux machines. Each device that is the same hardware will have similar configurations. The Ubuntu server is an Apache web server. The Kali Linux machine is used to penetration test (pen-test) the network.

Configuration of Routers

In this section, there will be three routers that will be configured and implemented in the topology. The router hardware is Cisco IOU L3 157-3 2018.

R1

1. Double click on R1 to enter the command prompt. Type enable at the prompt to enter privileged exec mode.
2. Enter config t in the terminal to enter global configuration mode. From here, there will be a few basic commands to set up the router. Type hostname R1 to change the hostname of the device to R1.
3. Type no ip domain-lookup to prevent IP Domain Name System hostname translation.
4. Type banner motd # Unauthorized access is strictly prohibited. # to set the login alert banner.
5. To set passwords, type security passwords min-length 15 to set the minimum length of passwords to 15 characters. Type service password encryption to encrypt system passwords. Type enable algorithm-type scrypt secret *password* to set the privileged exec

mode password and is encoded using the scrypt hashing algorithm.

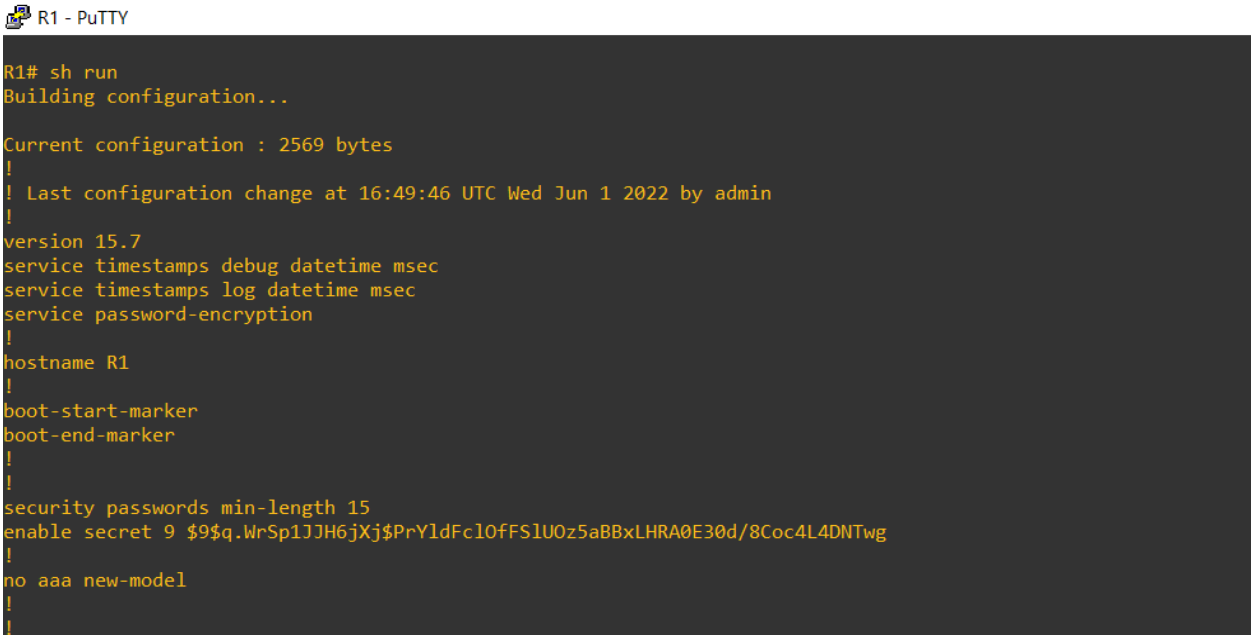


Figure 1 - sh run of hostname and security password

6. Next, create a user for the device. Type username *user* algorithm-type scrypt secret *password.* This creates a user to log on to the device. The password is hashed using the scrypt hashing algorithm.
7. Now, to set the passwords to be required at login. Type line vty 0 4 to configure the virtual terminal lines. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.
8. Next, type line con 0 to configure the primary terminal line. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.

9. Type line aux 0 to configure the auxiliary line. Type login local to enable local password checking. Exit the configuration.

```
line con 0
 exec-timeout 3 30
 privilege level 15
 logging synchronous
 login local
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 login local
line vty 0 4
 exec-timeout 3 30
 login local
 transport input none
!
!
end

R1(config)#
```

Figure 2 - sh run of line configurations

10. At the global configuration mode, type login block-for 120 attempts 5 within 60 to block failed login attempts for 2 minutes after 5 tries within 60 seconds.

11. Enter login on-failure log to generate syslogs of failed login attempts.



Figure 3 - sh run of login block-for

12. Enter interface eth0/0 at the global configuration mode to configure the Ethernet 0/0 interface. Enter ip address 192.168.10.1 255.255.255.192 (subnet mask) and then type no shutdown to activate the interface. Exit the configuration.
13. Enter interface eth0/1 at the global configuration mode to configure the Ethernet 0/1 interface. Enter ip address 192.168.20.1 255.255.255.0 and then type no shutdown to activate the interface. Exit the configuration.
14. Now for inter-VLAN routing. Enter interface eth0/0.10 at the global configuration mode to configure the Ethernet 0/0.10 sub interface. Type encapsulation dot1q to set the encapsulation type to IEEE 802.1Q VLAN. Enter ip address 192.168.10.65 0.0.0.13 (wildcard address) and then type no shutdown to activate the interface. Exit the

configuration.

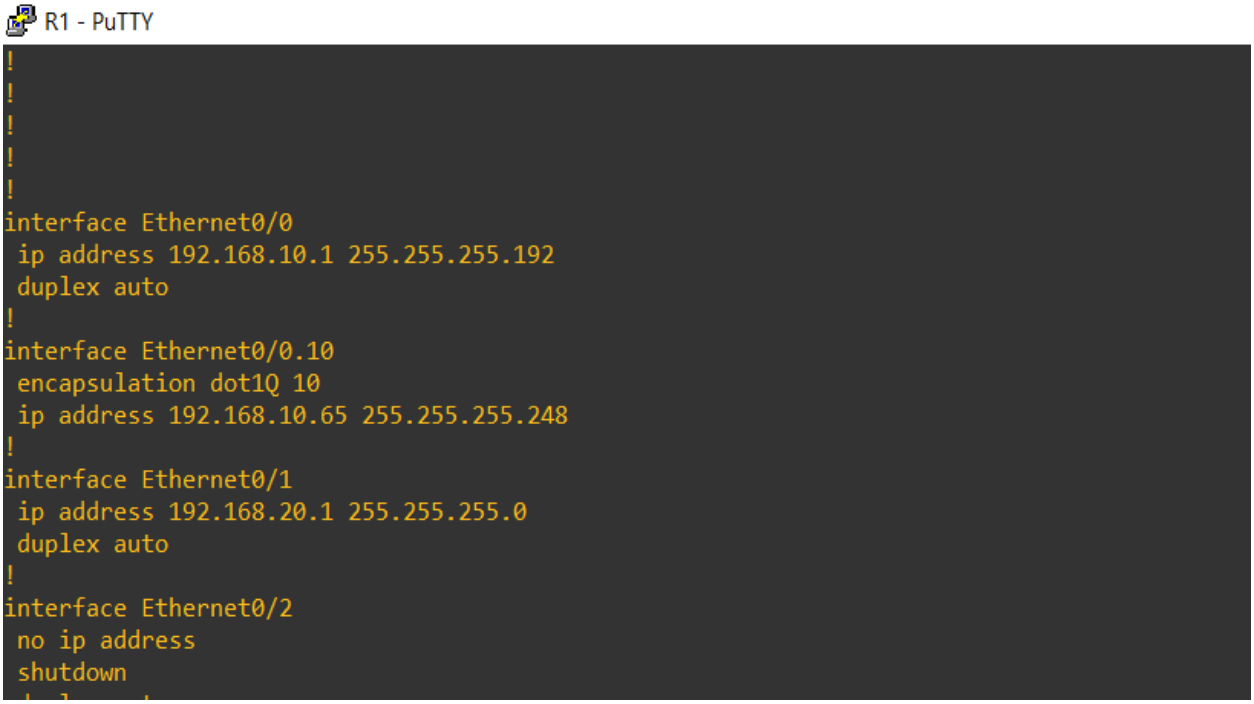


Figure 4 - sh run of interfaces

15. Shutdown all other interfaces by typing interface range eth0/2-3 then shutdown. Next, type interface range eth1/0-3 then shutdown. Type interface range serial 2/0-3 then shutdown. Type interface range serial 3/0-3 then shutdown. Exit the configuration.

16. To create a route to the other networks, type router ospf 100 at the global configuration mode. Type network 192.168.10.0 0.0.0.63 area 1 and type network 192.168.20.0 0.0.0.255 area 0. Then passive interface eth0/1. Exit the configuration.
17. To create a default static route, enter the global configuration mode. Then, type ip route 0.0.0.0 0.0.0.0 e0/0. Exit the configuration.

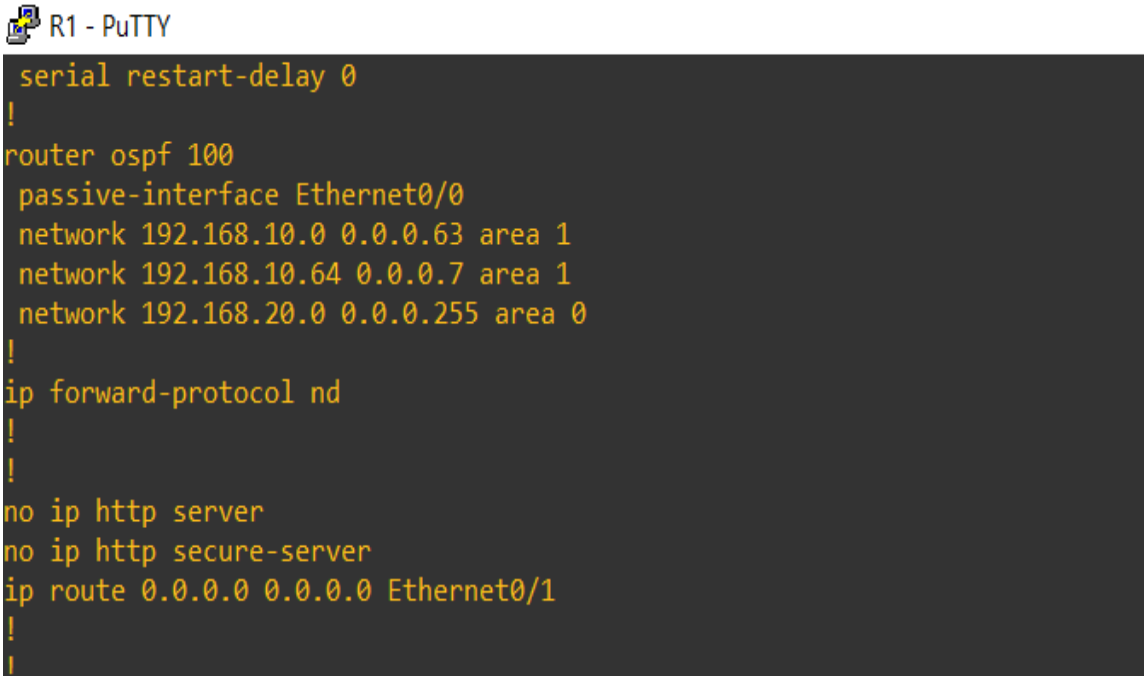


Figure 5 - sh run of ip routes on R1

18. Type copy running-config starting-config at the privileged exec mode to save the configuration to the startup configuration.

R2

1. Double click on R2 to enter the command prompt. Type enable at the prompt to enter privileged exec mode.

2. Enter config t in the terminal to enter global configuration mode. From here, there will be a few basic commands to set up the router. Type hostname R2 to change the hostname of the device to R2.
3. Type no ip domain-lookup to prevent IP Domain Name System hostname translation.
4. Type banner motd # Unauthorized access is strictly prohibited. # to set the login alert banner.
5. To set passwords, type security passwords min-length 15 to set the minimum length of passwords to 15 characters. Type service password encryption to encrypt system passwords. Type enable algorithm-type scrypt secret *password* to set the privileged exec mode password and is encoded using the scrypt hashing algorithm.

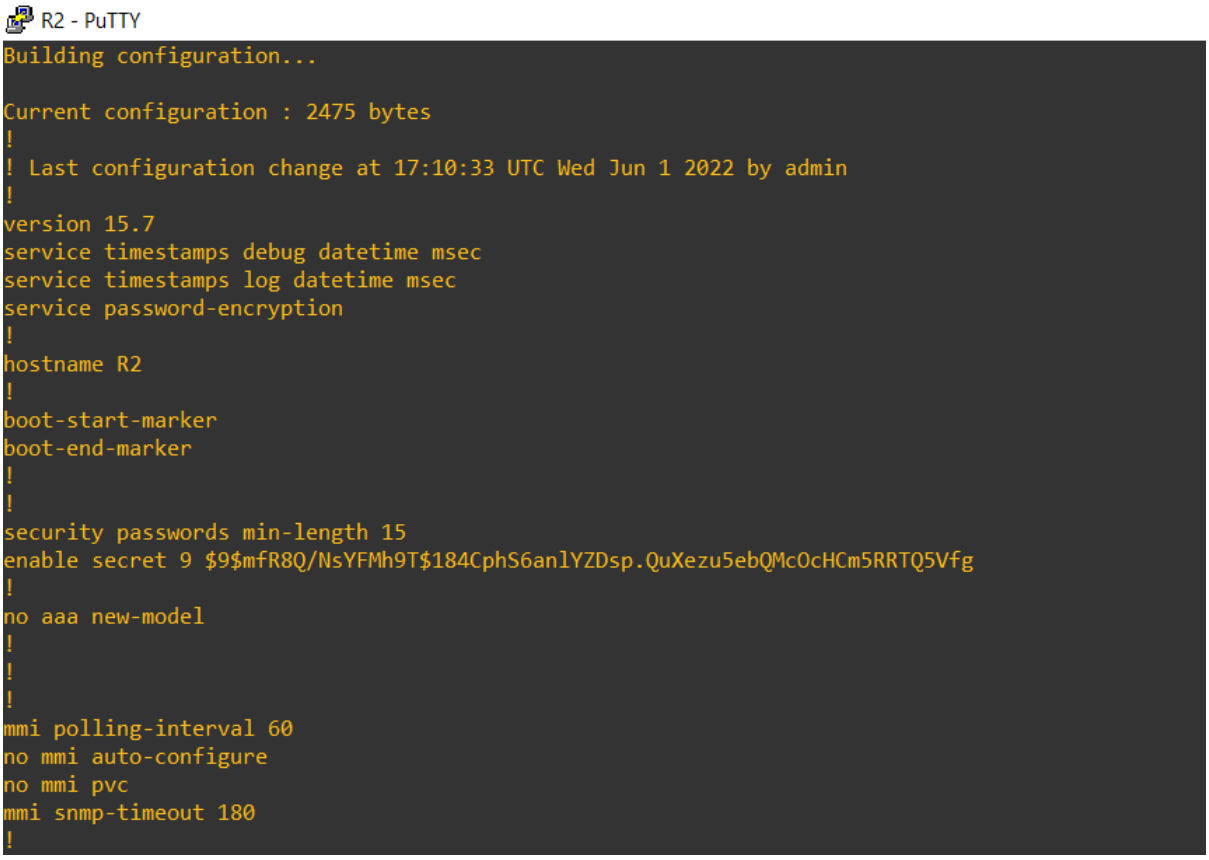


Figure 6 - sh run of hostname and security password

6. Next, create a user for the device. Type username *user* algorithm-type scrypt secret *password.* This creates a user to log on to the device at the global configuration mode. The password is hashed using the scrypt hashing algorithm.
7. Now, to set the passwords to be required at login. Type line vty 0 4 to configure the virtual terminal lines. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.
8. Next, type line con 0 to configure the primary terminal line. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.
9. Type line aux 0 to configure the auxiliary line. Type login local to enable local password checking. Exit the configuration.



Figure 7 - sh run of line configurations

10. At the global configuration mode, type login block-for 120 attempts 5 within 60 to block failed login attempts for 2 minutes after 5 tries within 60 seconds.

11. Enter login on-failure log to generate syslogs of failed login attempts.

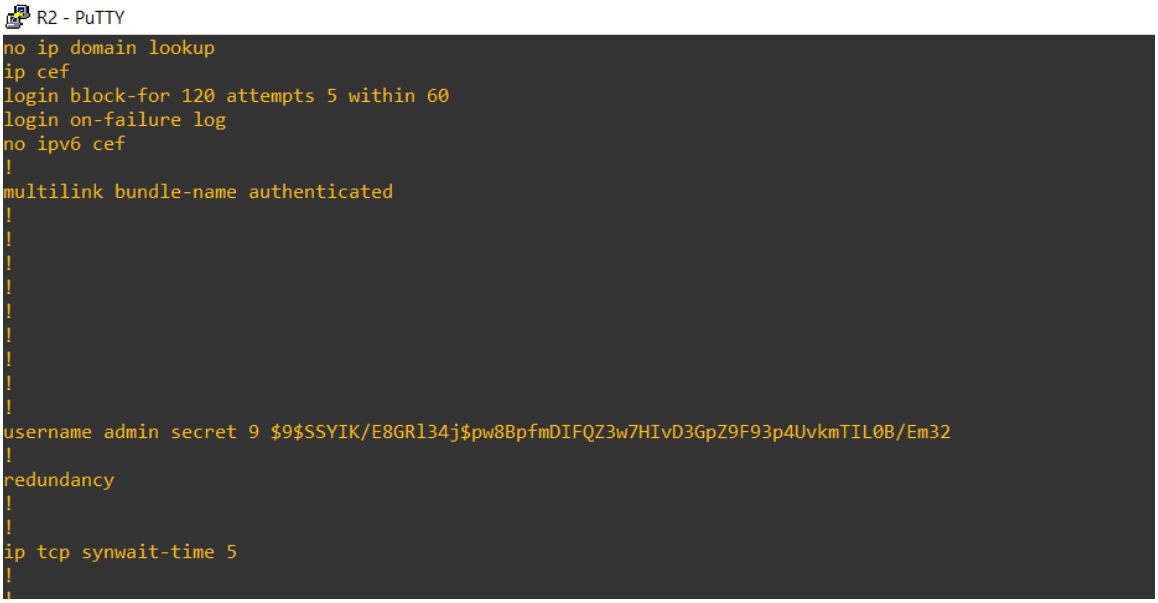


Figure 8 - login block-for in sh run

12. Enter interface eth0/0 at the global configuration mode to configure the Ethernet 0/0 interface. Enter ip address 192.168.20.1 255.255.255.0 (subnet mask) and then type no shutdown to activate the interface. Exit the configuration.
13. Enter interface eth0/1 at the global configuration mode to configure the Ethernet 0/1 interface. Enter ip address 192.168.30.1 255.255.255.0 and then type no shutdown to activate the interface. Exit the configuration.
14. Enter interface eth0/2 at the global configuration mode to configure the Ethernet 0/2 interface. Enter ip address dhcp. Note: The interface will be given a lease from the internet attached to the interface. Type no shutdown to activate the interface. Exit the

configuration.

R2 - PuTTY

```
!
!
!
!
!
interface Ethernet0/0
 ip address 192.168.20.2 255.255.255.0
 duplex auto
!
interface Ethernet0/1
 ip address 192.168.30.1 255.255.255.0
 duplex auto
!
interface Ethernet0/2
 no ip address
 shutdown
 duplex auto
!
```

Figure 9 - sh run for interfaces

15. Shutdown all other interfaces by typing interface range eth0/2-3 then shutdown. Next, type interface range eth1/0-1 then shutdown. Type interface range serial 2/0-3 then shutdown. Type interface range serial 3/0-3 then shutdown. Exit the configuration.
16. To create a route to the other networks, type router ospf 100 at the global configuration mode. Type network 192.168.20.0 0.0.0.255 area 0 and type network 192.168.30.0

0.0.0.255 area 0. Exit the configuration.

R2 - PuTTY

```
interface Serial3/3
 no ip address
 shutdown
 serial restart-delay 0
!
router ospf 100
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
```

Figure 10 - sh run router ospf

17. To create an access-list, enter the global configuration mode and type ip access-list extended 100. Enter permit ip 192.168.10.0 0.0.0.255 host 192.168.83.2. Then, enter permit ip 192.168.20.0 0.0.0.255 host 192.168.83.2. Next, enter permit ip 192.168.30.0 0.0.0.255 host 192.168.83.2. Finally, enter permit ip 192.168.40.0 0.0.0.255 host 192.168.83.2. The last statement will be an explicit deny ip any any. Exit the

configuration.

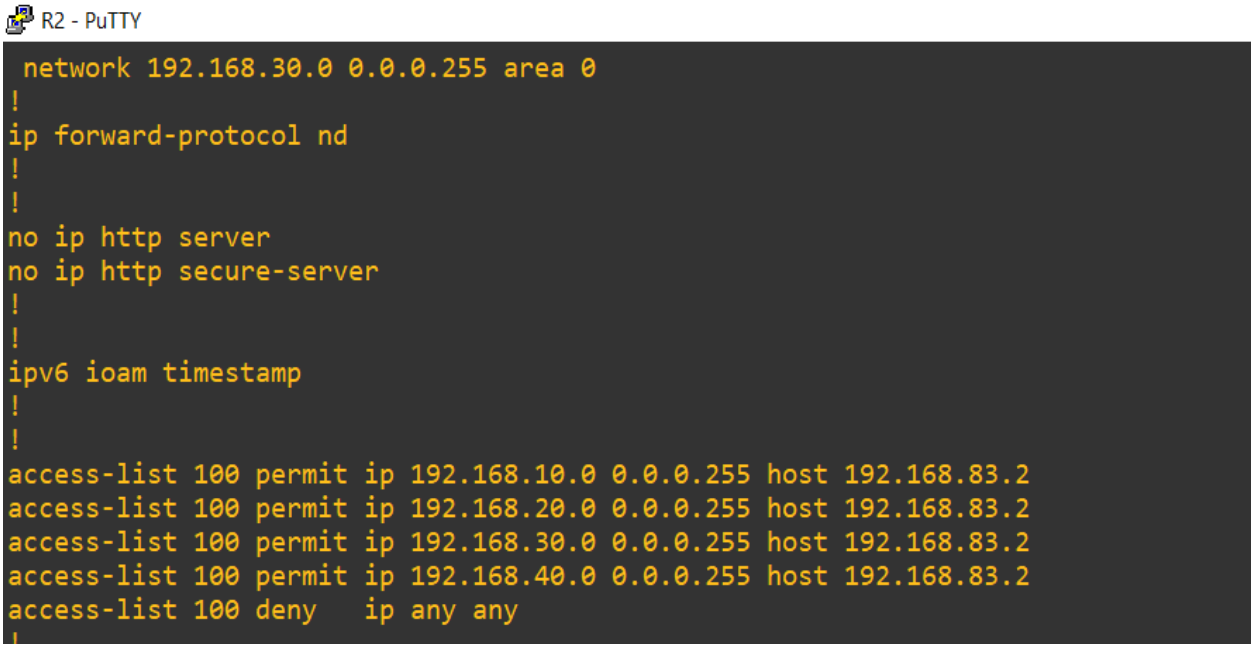


Figure 11 - sh run of access-list 100 on R2

18. Type copy running-config starting-config at the privileged exec mode to save the configuration to the startup configuration.

R3

1. Double click on R3 to enter the command prompt. Type enable at the prompt to enter privileged exec mode.
2. Enter config t in the terminal to enter the global configuration mode. From here, there will be a few basic commands to set up the router. Type hostname R1 to change the hostname of the device to R3.
3. Type no ip domain-lookup to prevent IP Domain Name System hostname translation.
4. Type banner motd # Unauthorized access is strictly prohibited. # to set the login alert banner.

5. To set passwords, type security passwords min-length 15 to set the minimum length of passwords to 15 characters. Type service password encryption to encrypt system passwords. Type enable algorithm-type scrypt secret *password* to set the privileged exec mode password and is encoded using the scrypt hashing algorithm.

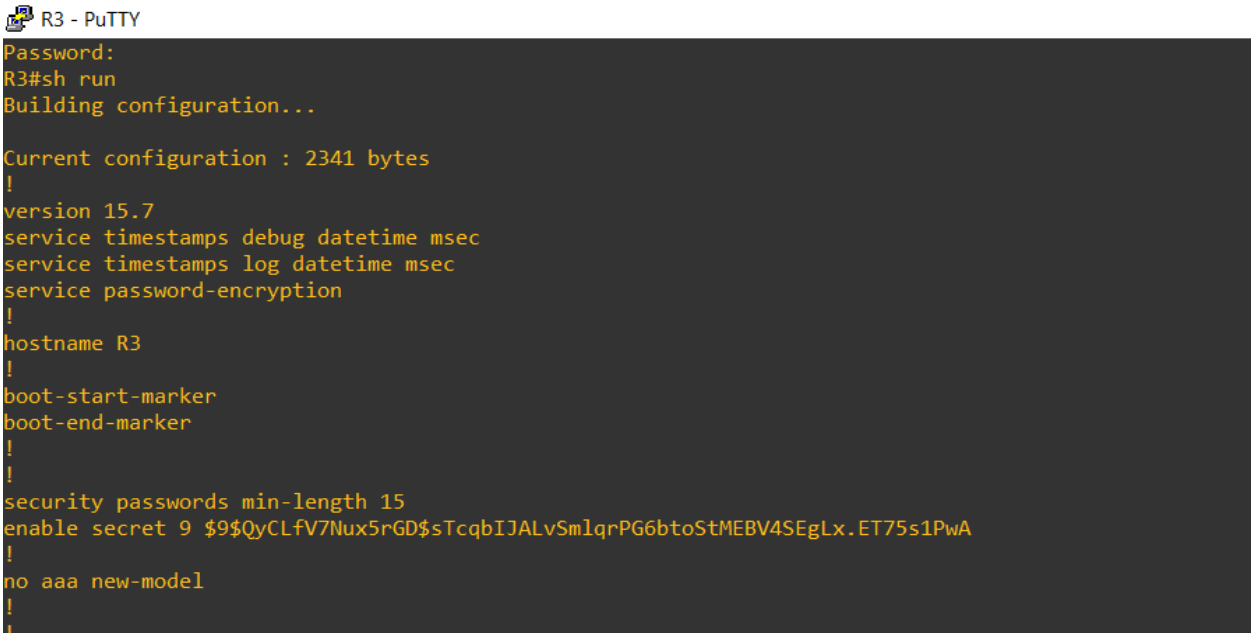


Figure 12 - sh run of hostname and security password

6. Next, create a user for the device. Type username *user* algorithm-type scrypt secret *password.* This creates a user to log on to the device at the global configuration mode. The password is hashed using the scrypt hashing algorithm.
7. Now, to set the passwords to be required at login. Type line vty 0 4 to configure the virtual terminal lines. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.
8. Next, type line con 0 to configure the primary terminal line. Enter exec-timeout 3 30. This will disconnect the user when there is 3 min and 30 seconds of inactivity. Type login local to enable local password checking.

9. Type line aux 0 to configure the auxiliary line. Type login local to enable local password checking. Exit the configuration.

```
!
!
banner motd ^C
Unauthorized access to this device is strictly prohibited.^C
!
line con 0
 exec-timeout 3 30
 privilege level 15
 logging synchronous
 login local
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 login local
line vty 0 4
 exec-timeout 3 30
 login local
 transport input none
!
!
end

R3#
```

Figure 13 - sh run of line configurations

10. At the global configuration mode, type login block-for 120 attempts 5 within 60 to block failed login attempts for 2 minutes after 5 tries within 60 seconds.

11. Enter login on-failure log to generate syslogs of failed login attempts.

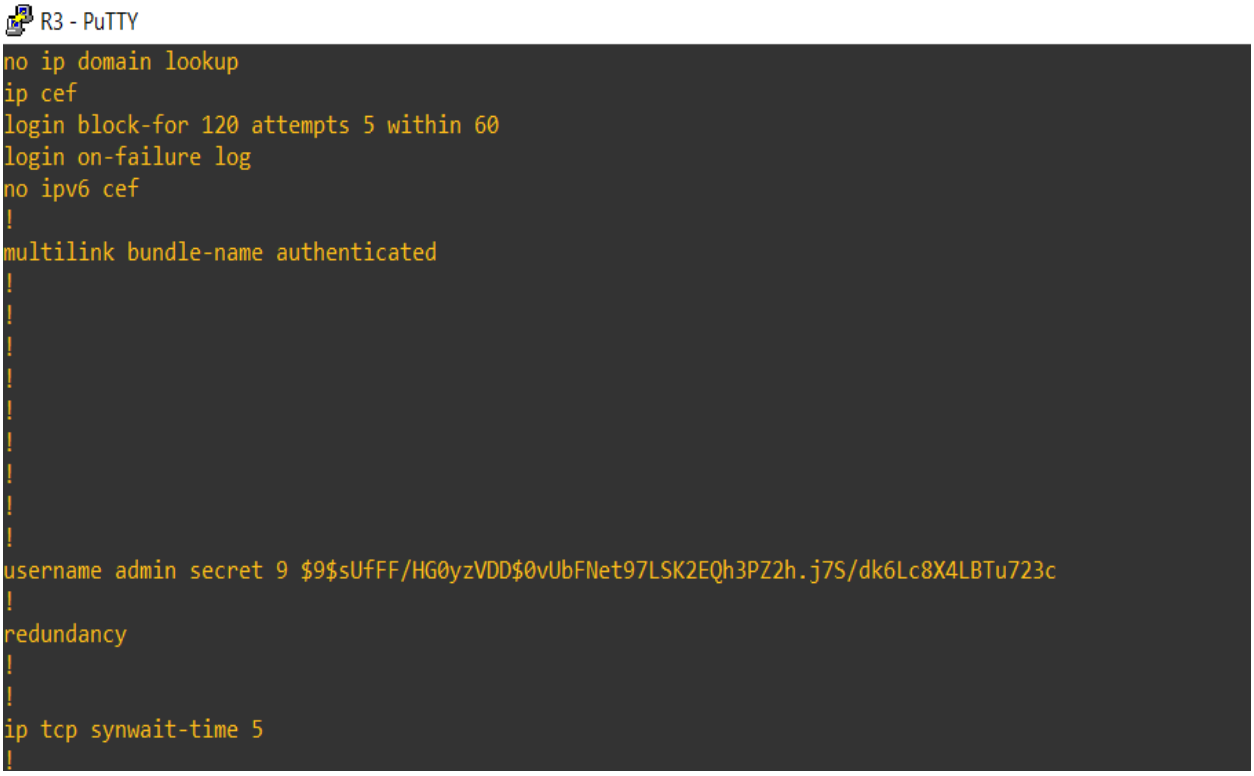


Figure 14 - login block-for in sh run

12. Enter interface eth0/0 at the global configuration mode to configure the Ethernet 0/0 interface. Enter ip address 192.168.20.1 255.255.255.0 (subnet mask) and then type no shutdown to activate the interface. Exit the configuration.

13. Enter interface eth0/1 at the global configuration mode to configure the Ethernet 0/1 interface. Enter ip address 192.168.30.1 255.255.255.0 and then type no shutdown to

activate the interface. Exit the configuration.

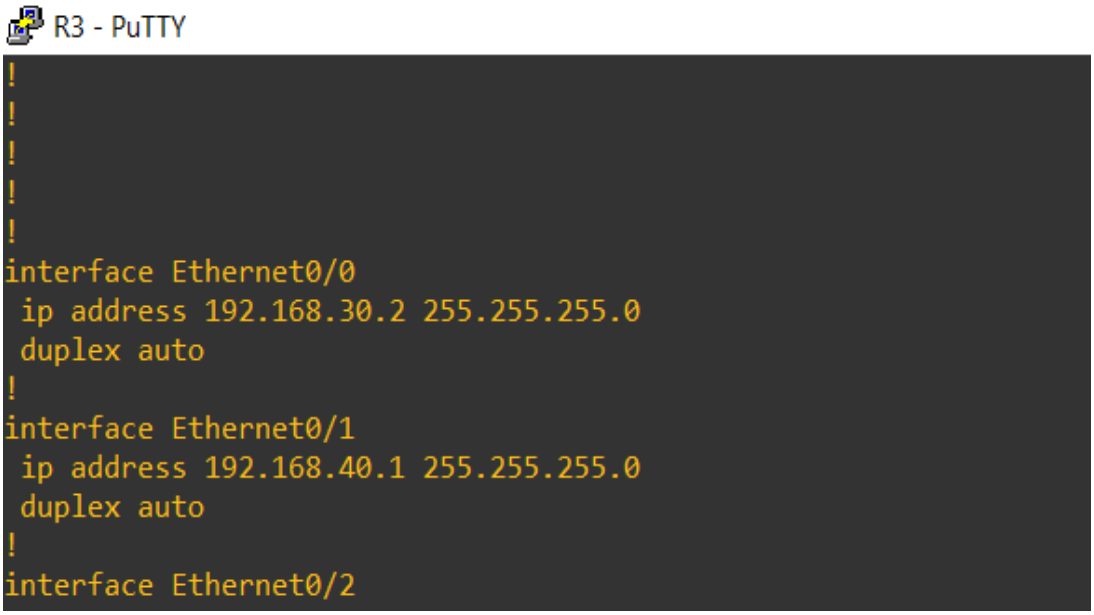


Figure 15 - sh run for interfaces

14. Shutdown all other interfaces by typing interface range eth0/2-3 then shutdown. Next, type interface range eth1/0-1 then shutdown. Type interface range serial 2/0-3 then shutdown. Type interface range serial 3/0-3 then shutdown. Exit the configuration.
15. To create a route to the other networks, type router ospf 100 at the global configuration mode. Type network 192.168.30.0 0.0.0.255 area 0 and type network 192.168.40.0

0.0.0.255 area 2. Then type passive interface eth0/1 to disable update out of the ethernet 0/1 interface. Exit the configuration.

16. To create a default static route, enter the global configuration mode. Then, type ip route 0.0.0.0 0.0.0.0 e0/0. Exit the configuration.

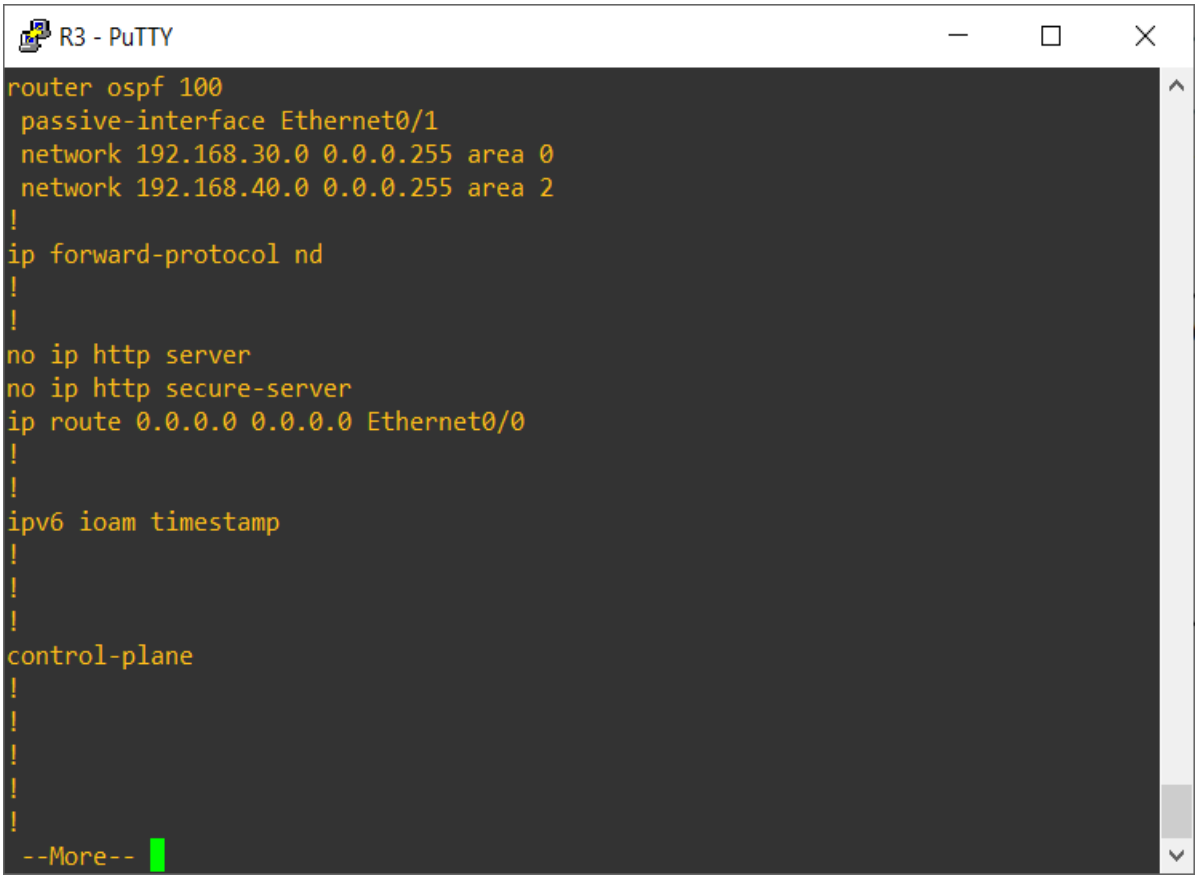


Figure 16 - sh run of R3 ip routes

17. Type copy running-config starting-config at the privileged exec mode to save the configuration to the startup configuration.

Switch Configuration

In this section, there will be one switch that is configured. The switch hardware is Cisco IOU L2 15.2 2019.

1. Double click on S1 to enter the command prompt. From here, type enable to enable privileged exec mode.
2. At the privileged exec mode, type config t to enter the global configuration mode.
3. Type hostname S1 to set the hostname of the device.
4. Enter service password-encryption to enable encryption of system passwords.
5. Type enable secret *password* to set the privileged exec mode secret password.

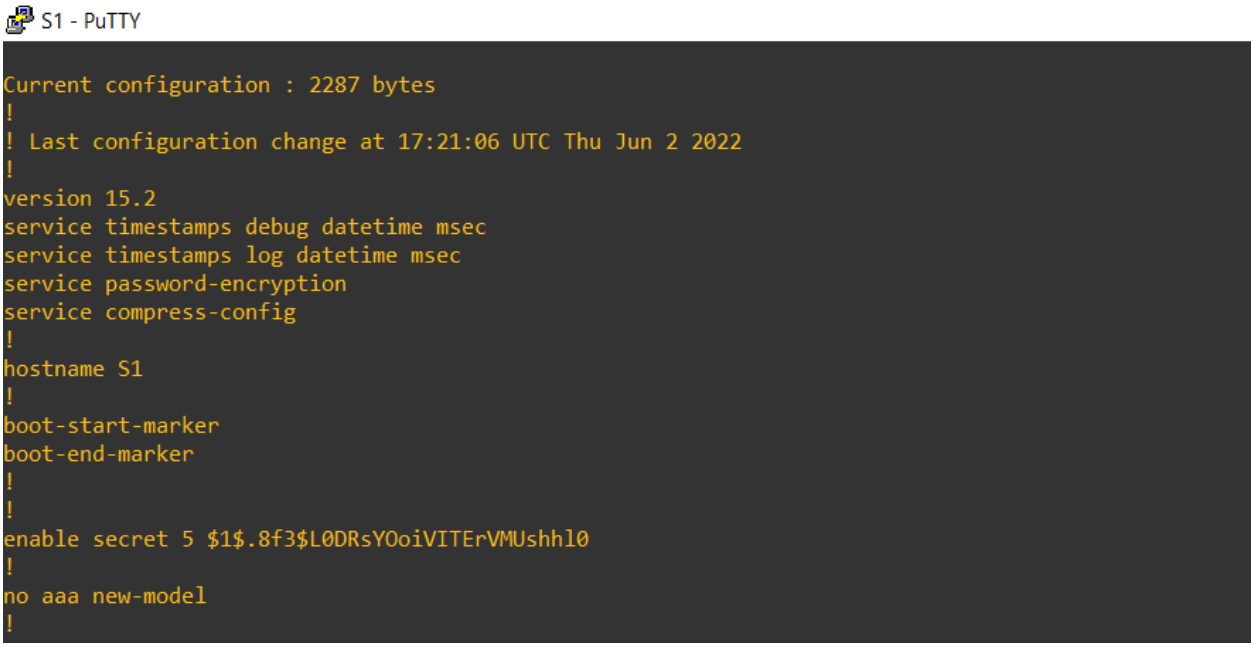


Figure 17 - sh run beginning of file

6. Enter line con 0 to configure the primary terminal line. Type password *password* to set the login password. Next, type exec-timeout 5 to automatically logoff after five minutes

of inactivity. Type login to enable password checking. Finally, type logging synchronous to enable synchronized message output of logs. Exit the configuration.

7. Enter line vty 0 4 to configure the virtual terminal line. Type password *password* to set the login password. Next, type exec-timeout 5 to automatically logoff after five minutes of inactivity. Type login to enable password checking. Finally, type logging synchronous to enable synchronized message output of logs. Exit the configuration.
8. Enter line aux 0 to configure the virtual terminal line. Type password *password* to set the login password. Next, type exec-timeout 5 to automatically logoff after five minutes of inactivity. Type login to enable password checking. Finally, type logging synchronous to enable synchronized message output of logs. Exit the configuration.
9. Type banner motd # Unauthorized access of this device is prohibited. # to set the alert prompt at the login screen.



Figure 18 - sh run of line configurations

10. At the global configuration mode, type interface vlan 10. Then, type ip address 192.168.10.66 255.255.255.248. Exit the configuration.

```
!
interface Vlan10
 ip address 192.168.10.66 255.255.255.248
!
```

Figure 19 - VLAN 10 configuration

11. Enter interface eth0/0 to configure the interface. Enter switchport mode access, then switchport access vlan 10 so the device connected will have access to VLAN 10. Enter switchport port-security maximum 2 to only allow a maximum of two secure addresses. Type switchport port-security mac-address sticky to configure dynamic secure addresses as sticky. Finally, type switchport port-security violation restrict to set the violation mode to restrict. Issue the no shutdown command to bring the interface up. Exit the configuration.
12. Enter interface eth0/1 to configure the interface. Enter switchport mode access, then switchport access vlan 10 so the device connected will have access to VLAN 10. Enter switchport port-security maximum 2 to only allow a maximum of two secure addresses. Type switchport port-security mac-address sticky to configure dynamic secure addresses as sticky. Finally, type switchport port-security violation restrict to set the violation mode to restrict. Issue the no shutdown command to bring the interface up. Exit the configuration.
13. Enter interface eth0/2 to configure the interface. Enter switchport mode access, then switchport access vlan 10 so the device connected will have access to VLAN 10. Enter switchport port-security maximum 2 to only allow a maximum of two secure addresses.

Type switchport port-security mac-address sticky to configure dynamic secure addresses as sticky. Finally, type switchport port-security violation restrict to set the violation mode to restrict. Issue the no shutdown command to bring the interface up. Exit the configuration.

14. Enter interface eth0/3 to configure the interface. Type switchport mode trunk to enable trunking for the interface. Next, type switchport trunk allowed vlan 10 to trunk the VLAN 10. Then, type switchport trunk encapsulation dot1q to enable IEEE 802.1Q encapsulation. Finally type switchport nonegotiate to disable negotiation protocol on this interface. Issue the no shutdown command to bring the interface up. Exit the configuration.

S1 - PuTTY

```
!
interface Ethernet0/0
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
!
interface Ethernet0/1
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
!
interface Ethernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
!
interface Ethernet0/3
 switchport trunk allowed vlan 10
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport nonegotiate
!
```

Figure 20 - interface configurations

15. Shut down all other interfaces by typing interface range eth1/0-3 then shutdown. Interface range eth2/0-3 then shutdown. And finally, interface range eth3/0-3 then shutdown.
16. At the privileged exec mode, type copy run start to save the running configuration to the startup configuration.

Computer Workstations

In this section, there will be two generic GNS3 PCs configured. There is no specific hardware involved; however, the software is PuTTY.

PC1

1. Double click on PC1 to enter the command prompt.
2. Type ip 192.168.10.67/29 192.168.10.65. The IP address of the computer will be set with the subnet mask and default gateway.

```
PC1> sh

NAME   IP/MASK              GATEWAY           MAC                LPORT  RHOST:PORT
PC1    192.168.10.67/29     192.168.10.65     00:50:79:66:68:00  20014  127.0.0.1:20015
       fe80::250:79ff:fe66:6800/64

PC1>
```

Figure 21 - sh command

3. Type save to save the configuration to the startup configuration.

PC2

1. Double click on PC1 to enter the command prompt.

2. Type ip 192.168.10.68/29 192.168.10.65. The IP address of the computer will be set with the subnet mask and default gateway.

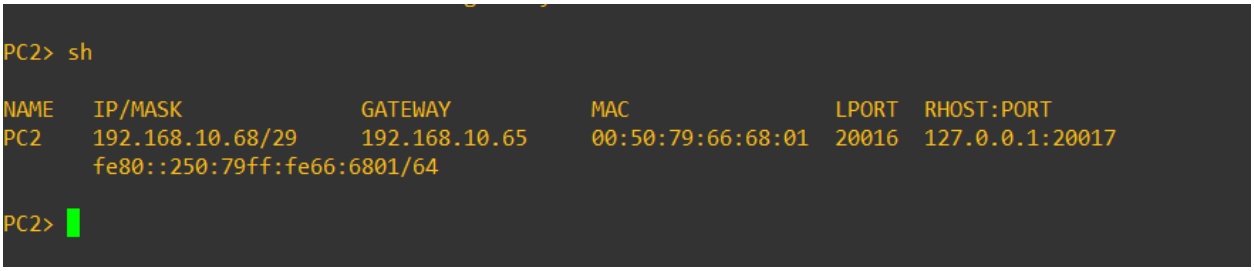


Figure 22 -sh command on PC2

3. Type save to save the configuration to the startup configuration.

Ubuntu Server Configuration

1. Double click on the Ubuntu Server. Type the password then press enter.
2. Click on the menu in the top left corner. Scroll down to System Tools then click MATE Terminal.
3. From here, type passwd to change the user password from the default password. Enter the current password, then the new password, and finally type the new password again. Press enter and the password will update.

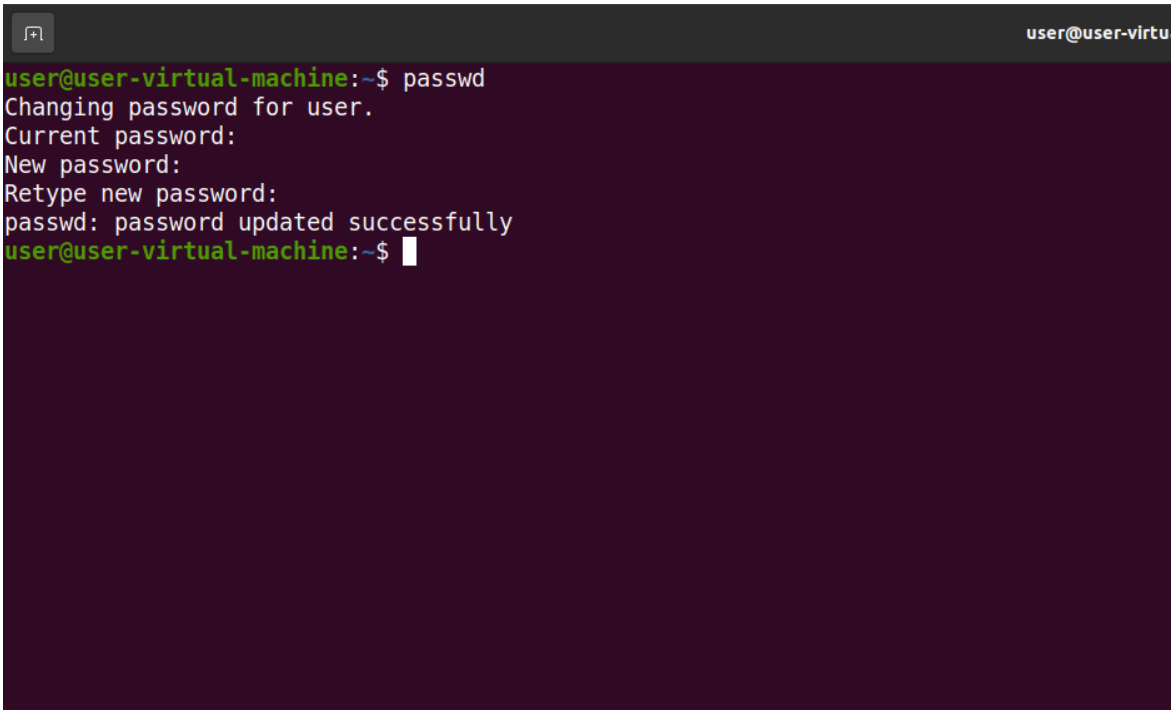


Figure 23 - setting the user password with the passwd command

4. In the top right corner, there is a Wi-Fi icon. Click the Wi-Fi icon to bring up the connections. Click on Edit Connections… then highlight the Wired connection 1. At the bottom of the prompt, there is a gear icon that brings up the settings. Click the gear icon.

5. In the Editing Wired connection 1 box, there are seven tabs at the top. Click on the IPv4 Settings tab.
6. Click the arrow next to the box labeled Method to bring up a menu. Select Manual for the method. To the right of the Addresses box, click the Add button. Type the address 192.168.40.100. Hit the tab button and keep the default netmask of 24. Click tab again and enter the gateway 192.168.40.1. Click save.

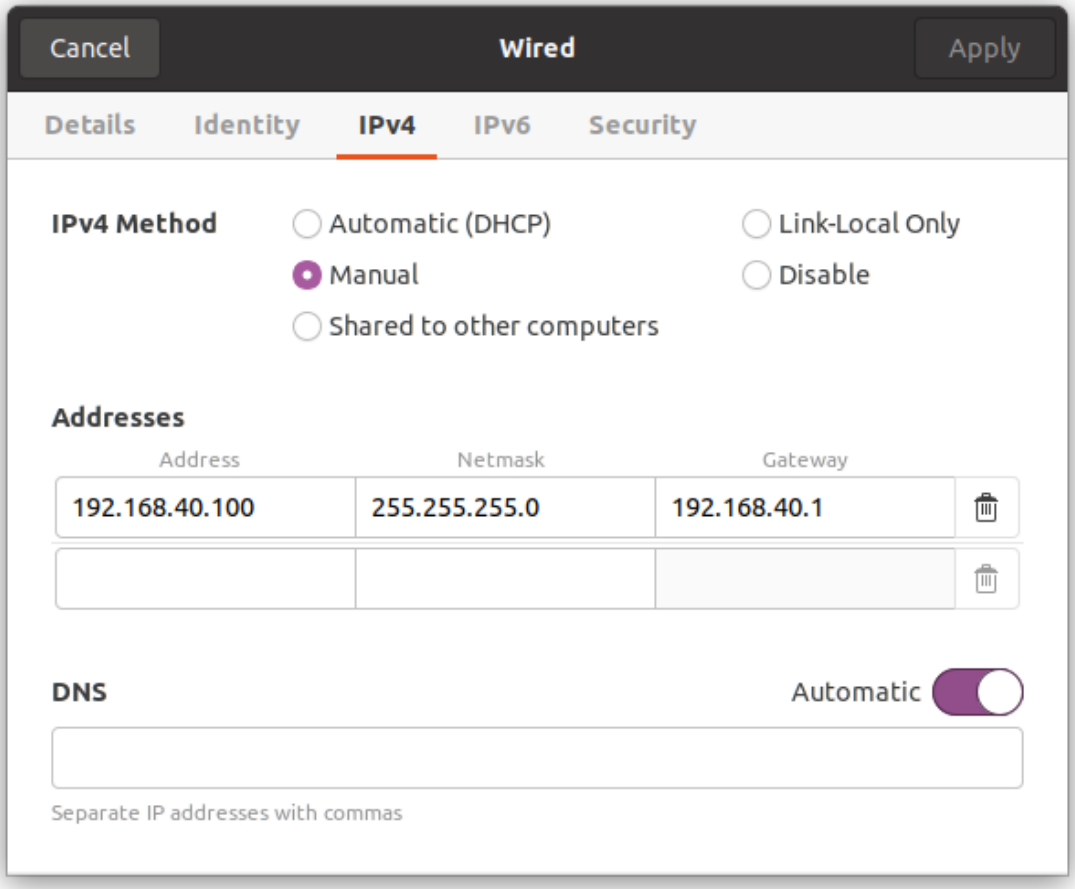


Figure 24 - editing the Wired connection 1 settings

7. After a few moments, a pop-up should display notifying of an established connection.
8. Download the apache2 service from the command line by typing sudo apt-get install apache2.

9. Create or download an html file for the webpage that is different from the default configuration.
10. After apache2 is successfully installed, start the service by typing at the command prompt service apache2 start.

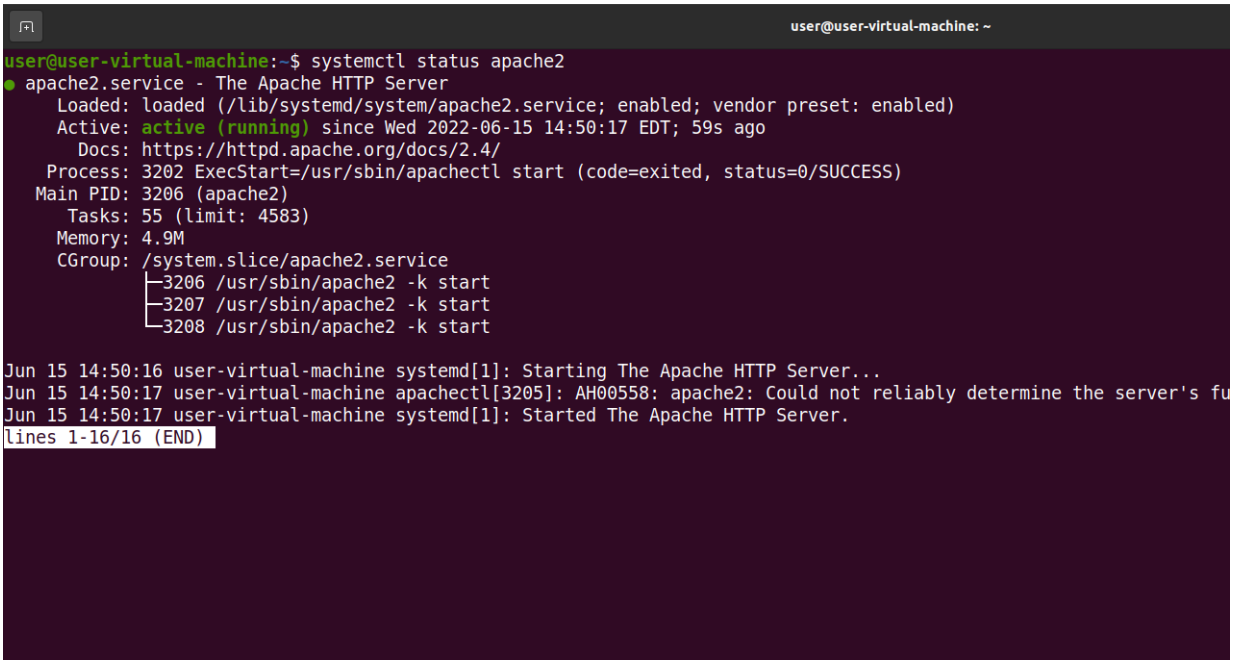


Figure 25 - systemctl status of apache2 service

11. Change the default webpage by going into the /var/www/html directory. From here, change the default index.html file to another name.
12. Move the new html file for the webpage to the /var/www/html directory by typing mv /home/user/downloads/example.html /var/www/html.
13. Inside the /var/www/html directory, change the new file to index.html by typing mv example.html index.html.

14. Navigate to the web browser and enter the IP address of the ubuntu server. The new webpage should load correctly.

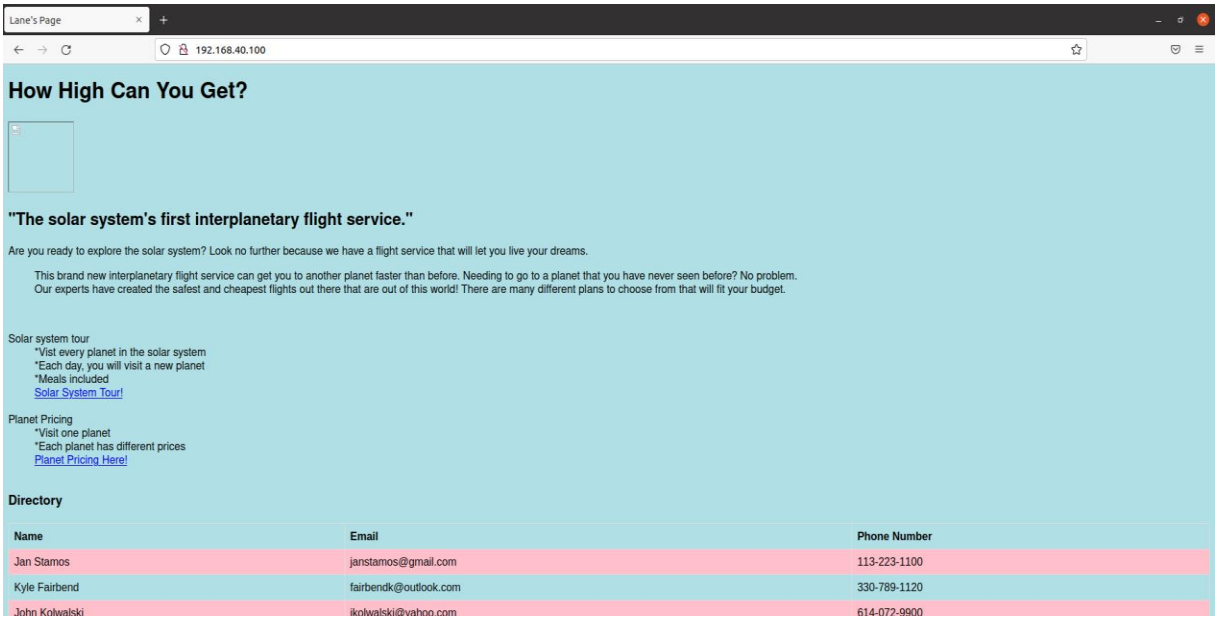


Figure 26 - webpage for 192.168.40.100

15. Download the VSFTP service by typing at the command line sudo apt-get install vsftpd.

16. After successfully installed, start the VSFTP service by typing service vsftpd start at the command prompt.

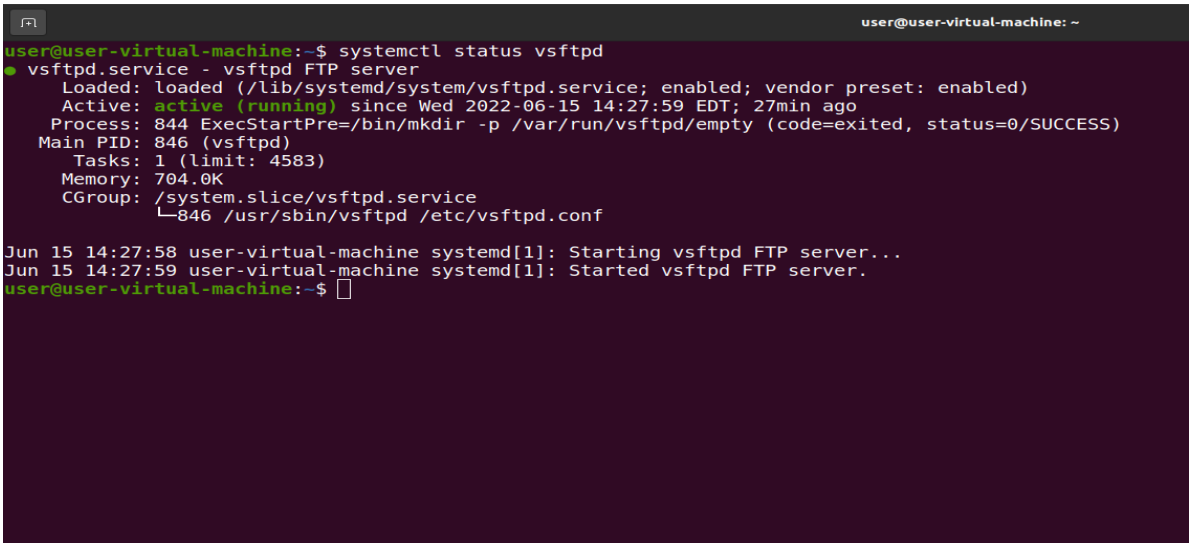


Figure 27 - systemctl status of vsftpd service

17. Download the SSH service by typing in sudo apt-get install ssh.

18. After successfully installed, start the SSH service by typing service ssh start at the command prompt.

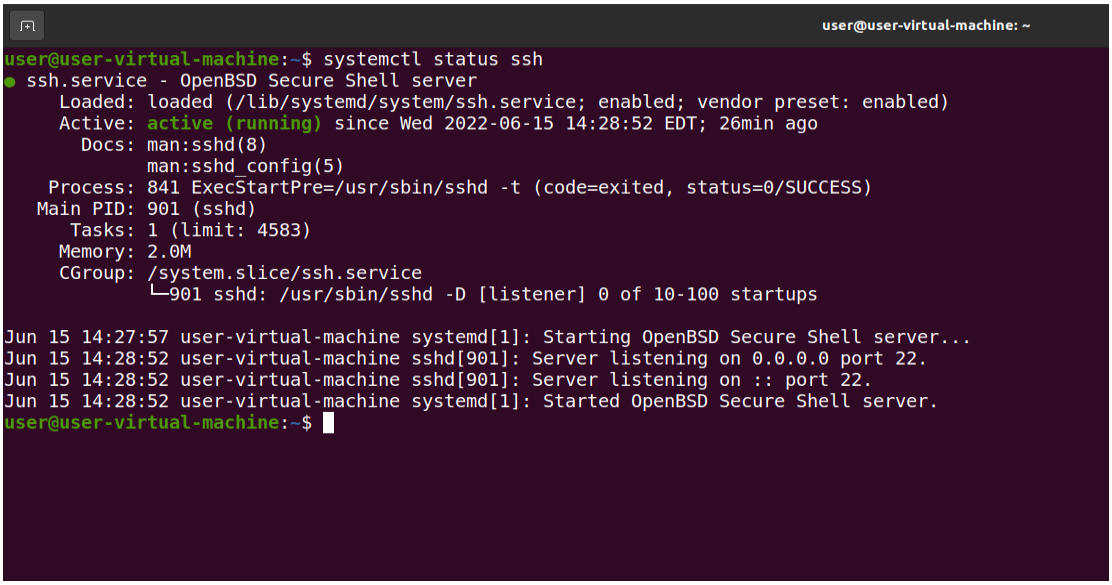


Figure 28 - systemctl status of ssh service

Kali Linux

In this section, there is one Kali Linux machine configured. The Kali Linux machine will pen-test the network by using tools already installed on the machine.

1. Double click on the Kali Linux machine and log in with the username and password.
2. At the home screen, click on the icon in the left corner. A drop-down menu will be displayed. Hover over Settings and click on Advanced Network Configuration.
3. Double click on Wired connection 1 to bring up the settings. Click on the IPv4 settings tab at the top. Change the method by clicking the box and changing from Automatic (DHCP) to Manual. Next, click Add button to the right of the Addresses box. From here, type the IP address 192.168.10.69, the Netmask 29, and the Gateway 192.168.10.65.
4. Click save to set the configurations and exit.

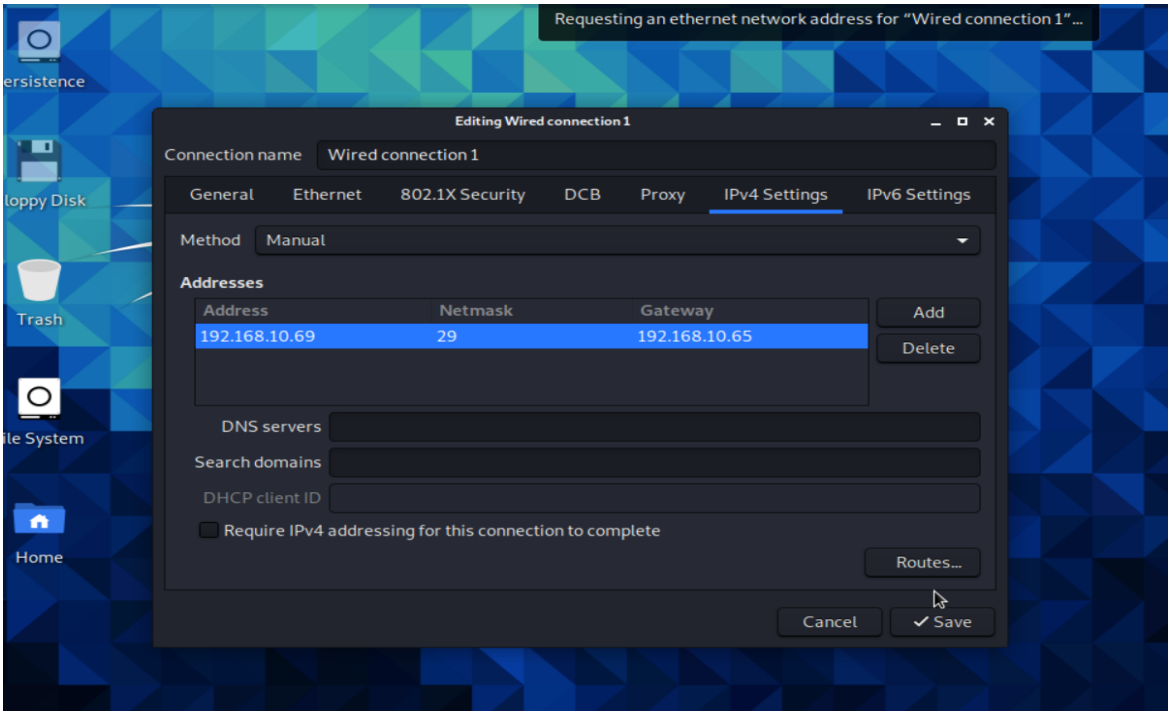


Figure 29 - Wired connection 1 settings

Nmap

Nmap is a command-line tool that is used to map the network. There are many features that are built in. The main feature that is used is the ability to scan a subnet.

1. On the Kali Linux machine, open the terminal.
2. Type sudo su root to change to the root user. Enter the password of the user.
3. At the command prompt enter nmap 192.168.40.0/24 to scan the subnet of R3.
4. In return, there should be open ports displayed for the Ubuntu server address.

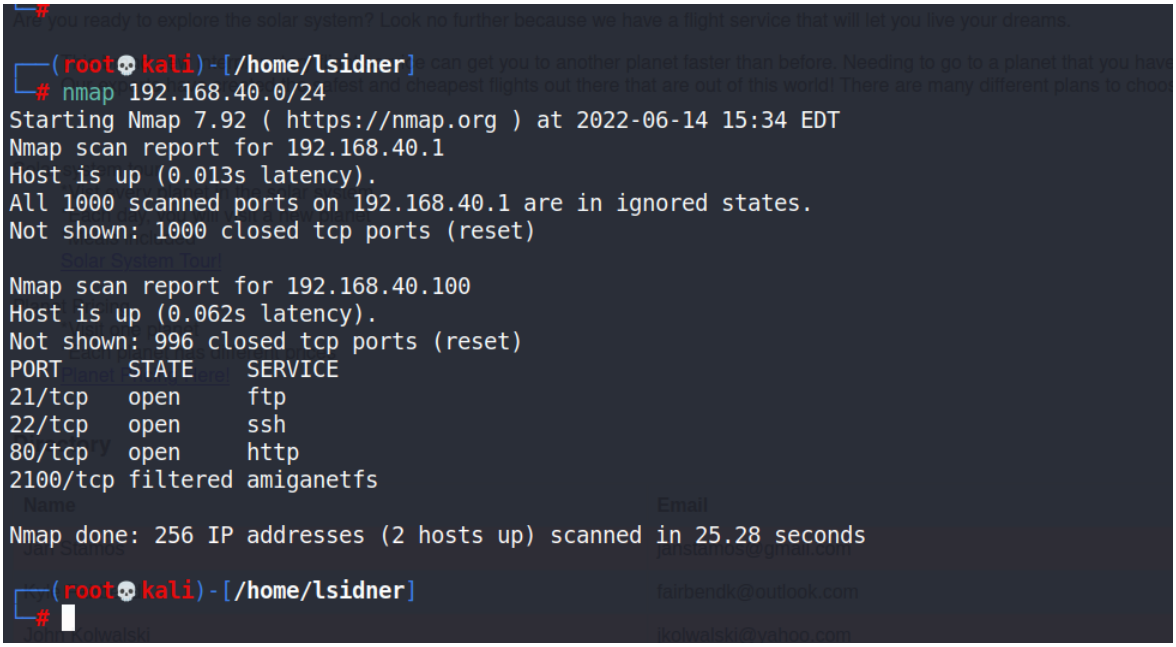


Figure 30 - nmap scan of 192.168.40.0/24

Metasploit

Metasploit is a pen-testing tool that is used from the command line. There are many tools that cna be used such as a port scanner. The tool that is focused on is the password sniffer known as psnuffle.

1. On the Kali Linux machine, open the terminal.
2. Type sudo su root to change to the root user. Enter the password of the user.
3. Type msfconsole at the command prompt to enter the Metasploit console.
4. Enter use auxiliary/sniffer/psnuffle at the command line.
5. Type run to start the password sniffer.
6. The output should display a log in to the Ubuntu server at 192.168.40.100. The username and password will be included in the log.

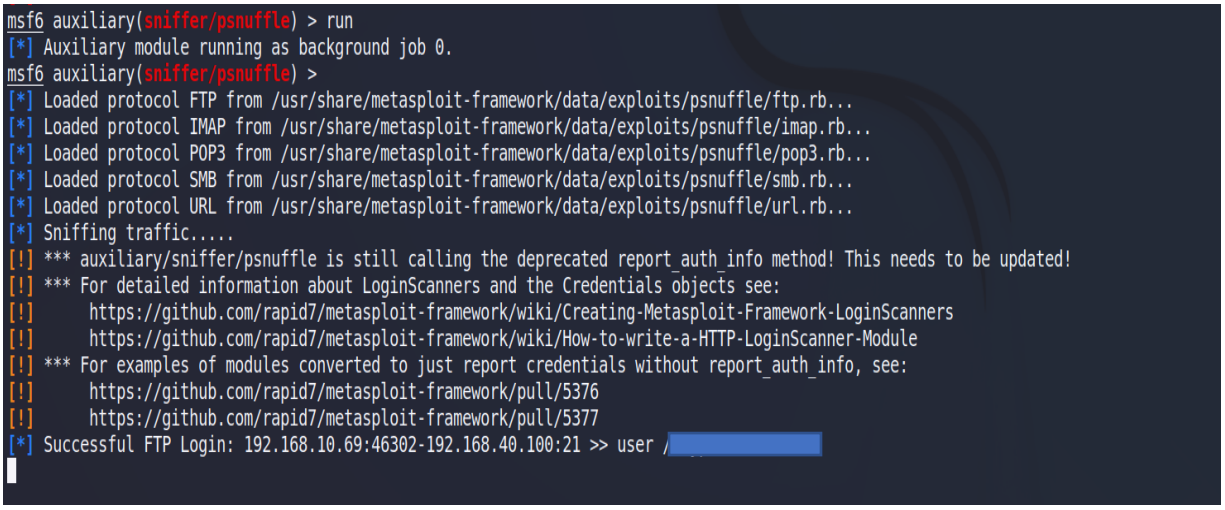


Figure 31 - auxiliary/sniffer/psnuffle output

John The Ripper

John the Ripper is used to crack password files for multiple operating systems such as Linux, macOS, and Windows. John the Ripper in this case is used to crack the shadow password file from the Ubuntu server.

1. On the Kali Linux machine, open the terminal.
2. Type sudo su root to change to the root user. Enter the password of the user.
3. Enter john *password_file.txt* to crack the password file.
4. The password for the username root will be displayed.

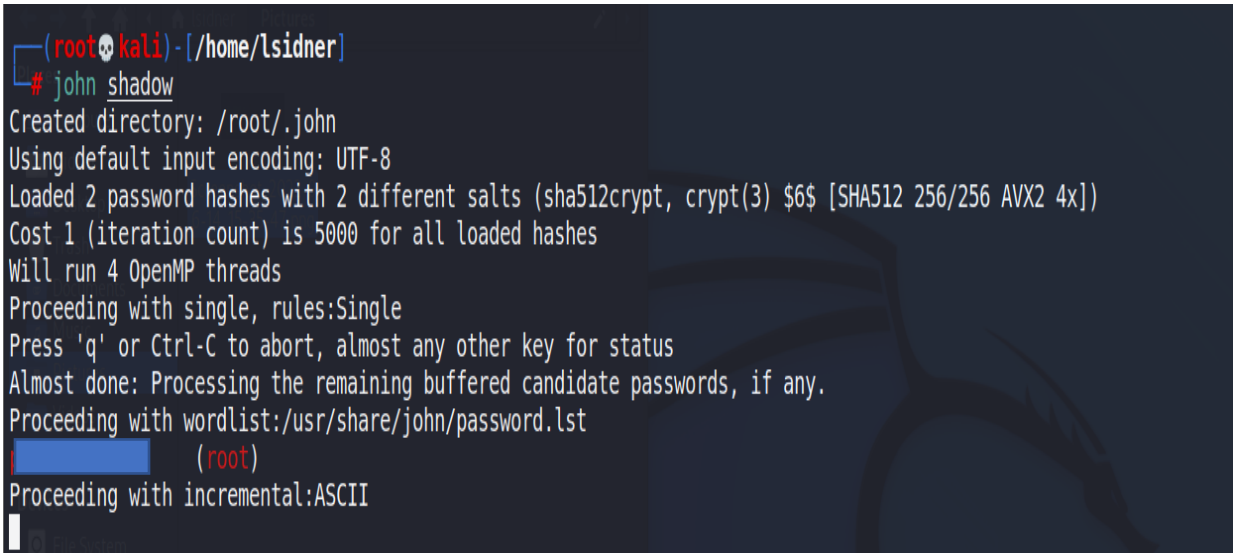


Figure 32 - john the ripper output